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EXPLAINABLE MACHINE LEARNING FOR TELECOM CUSTOMER CHURN

using XGBoost & SHAP

Section Number 2

Group 8

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ABSTRACT

This research addresses the critical challenge of customer churn prediction within the telecom industry, a crucial area for revenue protection and targeted marketing efforts. The study utilized a comprehensive dataset and employed the eXtreme Gradient Boosting (XGBoost) algorithm to develop a robust predictive model. A systematic methodology was implemented, focusing on meticulous data preprocessing using ColumnTransformer and addressing inherent class imbalance through the application of the scale_pos_weight parameter. Hyperparameter optimization was rigorously performed using Grid Search with 5-Fold Cross-Validation to identify the most generalizable model configuration. Upon evaluation on an unseen test set, the final XGBoost model demonstrated strong discriminatory power, achieving an AUC-ROC of 0.70. The study concluded with a detailed interpretation of the model using SHAP (SHapley Additive exPlanations) analysis, successfully identifying the key features driving customer churn predictions, thus providing actionable insights for strategic decision-making.

Index Terms- (keywords) Churn Prediction, Customer Retention, Explainable AI, Machine Learning, Predictive Modeling, Telecommunications.

1. INTRODUCTION

The telecoms industry is becoming increasingly competitive, making maintaining consumers a significant financial issue. Customer churn is when users cancel or switch services, which results in a considerable loss of revenue. As a result, precisely forecasting which consumers are likely to churn is critical, since it allows businesses to take proactive steps and offer targeted solutions to retain them. Machine learning (ML) offers powerful methods for analyzing vast and complicated usage and billing statistics, revealing hidden patterns that emerge before a consumer decides to leave. Therefore, the scope of this study is focused on the methodological development and internal validation of the predictive module using historical customer data, encompassing the entire process from data preprocessing through final performance evaluation.

2. PROBLEM DEFINITION AND MOTIVATION

The chosen problem focuses on building a predictive system that can identify customers who are at risk of churning, based on analyzing their complex historical data such as call records, internet usage, and complaints. The main motivation behind improving this system is its strong economic and industrial impact, since acquiring new customers is much more expensive than keeping an existing one. Machine learning and spatially advanced types (e.g. Random Forest or Ensemble methods) offers an effective solution by processing large amounts of customer data and identifying multiple factors that simple analysis cannot easily detect. This helps telecommunication companies allocate their retention efforts more efficiently. Read already published work in the same field.

3. LITERATURE REVIEW

3.1 Current State of the Art

The current research landscape in customer churn prediction within the telecommunications sector is predominantly characterized by a comparative analysis of a

wide array of machine learning algorithms, ranging from traditional statistical models to advanced ensemble and deep learning techniques [1, 2, 3, 4, 5].

A significant and consistent trend is the shift towards ensemble and boosting *methods* such as Random Forest, XGBoost, AdaBoost, and LightGBM which have demonstrated superior predictive accuracy, frequently exceeding 90% and even reaching up to 99% in optimized scenarios [1, 2, 5, 6]. This pursuit of high performance is paralleled by an intensive focus on data preprocessing, particularly to address the prevalent issue of class imbalance using sophisticated techniques like SMOTE (Synthetic Minority Oversampling Technique) often combined with ENN (Edited Nearest Neighbors) [1, 5, 7, 8]. Recent advancements also show a growing, though less frequent, exploration of deep learning architectures like Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN), promising exceptional accuracy with sufficient data [6, 8, 10]. Furthermore, an emergent and critical sub-field addresses model transparency through Explainable AI (XAI) tools such as SHAP and LIME, aiming to decipher the decision-making process of otherwise opaque "black-box" models [7, 9].

3.2 Comparative Analysis of Methodologies

A critical examination of predominant methodologies reveals a fundamental and recurring trade-off between predictive accuracy and model interpretability. Traditional and inherently interpretable models, such as Logistic Regression and Decision Trees, offer high transparency, allowing analysts to trace the logic behind each prediction [4, 7]. However, their predictive power is often limited, with accuracy typically plateauing around 80-86% [2, 4], making them less competitive for complex, high-stakes predictions. In stark contrast, advanced ensemble and boosting models (e.g., Random Forest, XGBoost, AdaBoost) have become the de facto standard for achieving state-of-the-art accuracy, consistently outperforming other categories [1, 2, 5, 6]. The principal drawback of this category is its "black-box" nature; while they indicate which features are important globally, they fail to explain how these features interact to produce a specific prediction for an individual customer, severely limiting actionable business insights [7, 9]. Deep Learning models represent the frontier for raw predictive performance, with studies like [8] reporting accuracy up to 98.42%. Yet, this comes at the highest cost in terms of computational complexity, substantial data requirements, and near-total opacity[8, 10].

Finally, a niche but vital category focuses on post-hoc interpretability using XAI tools [7, 9], which adds an explanatory layer to complex models but introduces additional computational overhead and does not replace the need for inherently understandable models in time-sensitive operational environments. This section is summarized in Table 1.

Model Category	Representative Algorithms (from Reviewed Studies)	Key Strengths / Advantages	Key Weaknesses / Limitations	Typical Reported Accuracy (from Studies)
Traditional / Interpretable Models	Logistic Regression [2,4,7], Decision Trees [1,4,6], Naive Bayes [2]	• **High Interpretability &Transparency** (easy to trace decisions) [4,7]. • Lower computational cost & faster training. • Less prone to overfitting on small datasets.	• **Lower predictive accuracy** compared to complex models [2,4]. • Poor performance with non-linear or complex data patterns. • May require extensive feature engineering.	~77% - 86% (e.g., 80.45% for LR [2], 83% AUC for DT [2], 78% for DT before balancing [1])
Advanced Ensemble & Boosting Models	Random Forest [1,2,4,5,6], XGBoost [2,3,4,6], AdaBoost [2,6], LightGBM [6]	• **Highest Predictive Accuracy** (state-of-the-art performance) [1,2,5,6]. • Robust to overfitting and handle non-linear relationships well. • Provides feature importance scores.	• **"Black-Box" Nature** – low interpretability for specific predictions [7,9]. • Higher computational cost & memory usage. • Risk of overfitting if hyperparameters are not tuned properly.	**~90% - 99%** (e.g., 99.09% for RF [1], 81.71%-84% AUC for AdaBoost/XGBoost [2], 90.9% for RF [5])

Deep Learning Models	Artificial Neural Networks (ANN) [6,10], Convolutional Neural Networks (CNN) [8]	• **Potential for exceptional accuracy** with large, complex datasets [8,10]. • Automatic feature extraction from raw data. • Can model intricate, non-linear patterns.	• **Extremely Low Interpretability** (complex black-box) [8,10]. • **High computational complexity** & resource demands [8,10]. • Requires very large datasets for effective training [10].	**~93% - 98.4%** (e.g., 93.88% for ANN [10], 98.42% for CNN [8], 79%-80% for ANN [6])
Models with Explainable AI (XAI) Focus	Models explained via SHAP/LIME [7,9], Hybrid Interpretable Models	• **Improved model transparency** & insight into prediction reasons [7,9]. • Builds trust with business stakeholders. • Helps identify key drivers of churn for actionable strategies.	• **Adds computational overhead** to the modeling process [9]. • Explanations can be **local** (per instance) and not globally consistent. • Still relies on a base complex model that is itself a black box.	**~81% - 91.7%** (e.g., 81% accuracy with GBM+SHAP [9], 91.66% accuracy for RF with LIME/SHAP [7])

Table 1: Comparison of Machine Learning Algorithms used in current Literatures

3.2.1 Gap Analysis and Justification

A critical synthesis of the reviewed literature reveals three persistent and interconnected research gaps that limit the practical application of ML-driven churn prediction systems in the telecommunications sector.

- The Accuracy-Interpretability Trade-off in Advanced Models

A predominant trend in current research is the pursuit of maximizing predictive accuracy, often achieved through complex ensemble or deep learning models such as XGBoost, Random Forest, and Artificial Neurall Networks [1, 2, 5, 6]. However, this focus comes at a significant cost: model opacity. Studies that achieve high accuracy (e.g., 99% in [1]) simultaneously demonstrate the "black-box" nature of these models, offering little insight into the reasons behind individual churn predictions. For instance, research [1] necessitated a separate statistical model (Cox Proportional Hazard) for interpretability, while [3] conflated feature importance with true model explainability. This gap highlights a critical disconnect between technical performance and business utility, as retention strategies require actionable reasons, not just binary predictions.

- Data Dependency and Generalization Challenges

The performance and reported efficacy of many proposed models are heavily contingent on specific data conditions, raising concerns about robustness and real-world applicability. Key sub-gaps include:

  - Class Imbalance: The recurring issue of imbalanced datasets is frequently cited [7, 8, 10], leading to biased models that poorly predict the minority churn class.

  - Synthetic Data Reliance: To address imbalance, studies extensively depend on synthetic oversampling techniques like SMOTE and ENN [1, 5, 7, 8]. This reliance may create models tailored to artificial data distributions, potentially undermining their performance on genuine, unseen customer data.

  - Limited Dataset Scope and Size: Models are often trained and validated on single, specific datasets (e.g., the IBM Telco dataset) [6, 9], or datasets of limited size [4], which question their generalizability across different telecom markets and customer bases.

- Computational and Methodological Complexity for Practical Deployment

Emerging sophisticated methods, while theoretically powerful, introduce practical barriers to implementation. The computational complexity of certain ensemble methods and advanced feature selection algorithms (e.g., Gravitational Search Algorithm) is noted as a potential hindrance for real-time or large-scale deployment [6, 9]. Furthermore, some methodologies employ an extensive array of preprocessing and balancing techniques without a clear, streamlined rationale [5], suggesting a focus on experimental optimization over parsimonious, efficient pipeline design suitable for operational environments.

3.2.2 Justification for Proposed Work

The identified gaps, particularly the first concerning interpretability, create a significant opportunity for a more balanced ML module design. Merely achieving high accuracy is insufficient if the model's decisions cannot be trusted or understood by business stakeholders. Therefore, this review justifies the need for a novel ML module proposal that prioritizes bridging the accuracy-interpretability divide.

The proposed module will aim to incorporate explainability as a core design principle, potentially through hybrid or inherently interpretable model architectures, without substantially compromising predictive power, thereby delivering a tool that is both intelligent and actionable for telecom customer retention teams.

4. DETAILED ML MODULE

4.1 Objectives

The purpose of this module is to improve the performance and capability of the exiting churn prediction system.

The module Purposes to achieve the following objectives:

1- Improve Prediction Accuracy and Build Churn Prediction System That Achieves High Accuracy While Avoiding Issues Caused by Class Imbalance and Data Irregularities

2- Enhance Interpretability and Provide Transparent, Instance-Level Explanations for Each Prediction-Bridging the Accuracy-Interpretability Gap Identified in the literature.
3- Deliver Actionable Insights of the Churn Probabilities to Support Retention and Decision-Making Process.
4- Integrate Seamlessly into the System Pipeline to Enable Continuous Predictions and Insights for CRM Applications.

4.2 Approach

4.2.1 Selected Technique: Xgboost Classifier

<u>XGBoost ended up being selected because it:</u>

- Handels large and complex data sets competently.
- Helps automatic handling of missing values.
- Accomplishes strong performance in churn prediction problems.
- Operates smoothly with both numerical and categorical features.

4.2.2 Data Preprocessing and Feature Engineering

The raw dataset, sourced from a Telecom customer database [11], was loaded and initially prepared. The target was 'churn'. The unique identifier **'*Customer_ID*'** was dropped.

The dataset features were categorized into two main types:

- <u>Numerical Features:</u> Features containing continuous or discrete numerical values.
- <u>Categorical Features:</u> Features containing non-numeric, descriptive labels (type object).

4.2.2.1 Preprocessing Pipelines using Column Transformer

To ensure consistent and repeatable data transformation, preprocessing steps are contained within scikit-learn pipelines and organized using the

columnTransformer. This design allows different transformations to be applied to various feature groups at once.

The numerical features pipeline was processed through a dedicated pipeline highlighting and focusing on handling missing values by SimpleImputer with the median strategy was used.

The categorical features pipeline was processed through two steps first by using the SimpleImputer with constant strategy was applied to handling the missing values and One-Hot Encoding was applied to convert categorical values to numerical values.

4.2.2.2 Data Structuring and Final Splitting

The final step involved splitting the processed data into training and testing sets they were split into 80% for training and 20% for testing.

4.2.3 Training Methodology and Data Splitting

The training was performed using the pre-processed feature set ($\boldsymbol{x_{train}}$) and the target variable ($\boldsymbol{y_{train}}$)

- **Training Set:** Used for model fitting.
- **Validation:** Hyperparameter tuning was performed using 5-Fold Cross-Validation (CV=5) applied directly within the grid search mechanism to ensure strength and generalizability of the chosen parameters.
- **Evaluation Metric:** The Area Under the ROC Curve (AUC-ROC) was selected as the primary scoring metric. AUC-ROC is especially effective for imbalanced datasets, as it evaluates the model's ability to differentiate between classes throughout all classification thresholds, independent of the class distribution.

4.2.4 *Handling Class Imbalance (Weighting)*

The dataset explained significant class imbalance, where the number of negative samples (Non-churn, 0) considerably exceeded in the positive samples (Churn, 1). To mitigate the bias towards the majority class and improve the detection of the minority class, the ***scale_pos_weight*** hyperparameter was applied.

The weight was calculated as the ratio of negative to positive instances:

$$Scale\ pos\ weight = \frac{Count\ of\ Negative\ Samples}{Count\ of\ Positive\ Samples}$$

This calculated weight was fixed in the base XGBoost model to suitably penalize errors on the underrepresented positive class during training.

4.2.5 *Hyperparameter optimizing using GridSearchCV*

Optimal performance was accomplished by structured searching for the Most efficient combination of hyperparameters using **Grid Search Cross-Validation (GridSearchCV).**

The final search space was carefully defined to promote simple, robust trees while ensuring satisfactory learning capacity as shown in Table 2:

Hyperparameter	Description	Search Space / Fixed Value
max_depth	Maximum depth of a tree.	[3, 4, 5]
n_estimators	Number of boosting rounds (trees).	500
learning_rate	Step size shrinkage used in updating weights.	[0.05, 0.1]
subsample	Fraction of samples used for fitting base learners.	[0.7, 0.8]
colsample_bytree	Fraction of columns (features) used per tree.	[0.7, 0.8]

Table 2: Hyperparameter Search Space and Values for XGBoost Optimization

The best parameter set identified by GridSearchCV was:

colsample_bytree:0.7 , learning_rate: 0.05 , max_depth : 5 , subsample: 0.7.

4.2.6 Evaluation and performance validation

The performance of the best trained model was evaluated on test set to ensure its generalization ability on new customer data.

The performance metrics were used to evaluate the model power on predicting the churn or non-churn classes. AUC-ROC was chosen as the primary metric, along with accuracy and classification report (recall(sensitivity), precision and F1 score).

And for visualization we used AUC-ROC Curve as shown in **Figure 2** and Confusion matrix as shown in **Figure 3.**

4.2.6.1 Results Summary

The best-trained model (derived from the optimal hyperparameters in the search space) was applied to the Test Dataset. The results are summarized below in **Table 3:**

A. Performance Metrics on Test Set

Metric	Value	Interpretation
AUC-ROC	*[0.7]*	Indicates the model's strong discriminatory power.
F1-Score	*[0.64]*	Represents the balance between Precision and Recall.
Recall (Sensitivity)	*[0.65]*	Shows the percentage of actual Churn cases correctly caught.
Precision	*[0.63]*	Shows the reliability of the model's Churn predictions.
Accuracy	*[0.64]*	Shows the overall percentage of the correct predictions made by the model

Table 3 :Performance Metrics for XGBoost model

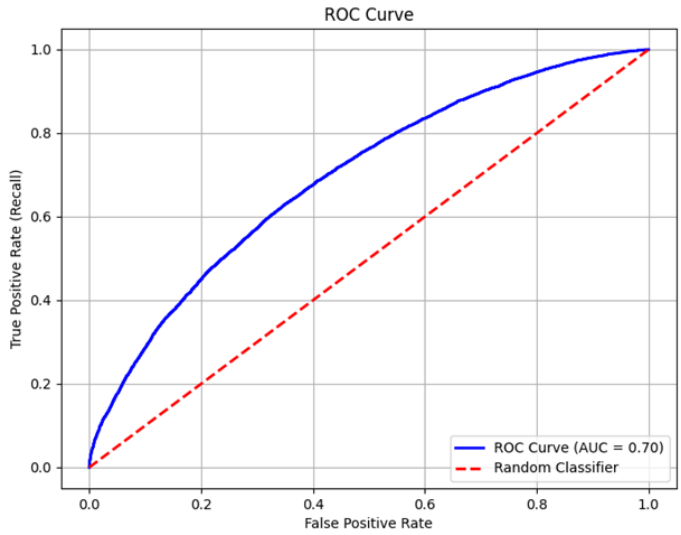

Figure 1: AUC-ROC Curve for XGBoost model

B. Confusion Matrix Analysis

Figure 2 visualizes the Confusion Matrix for the Test Set; it was analyzed to identify critical errors with business impact:

- False negative (FN): actual churns were missed by the model and predicted as non-churners
- False positive (FP): non-churners were missed by the model and predicted as churners.

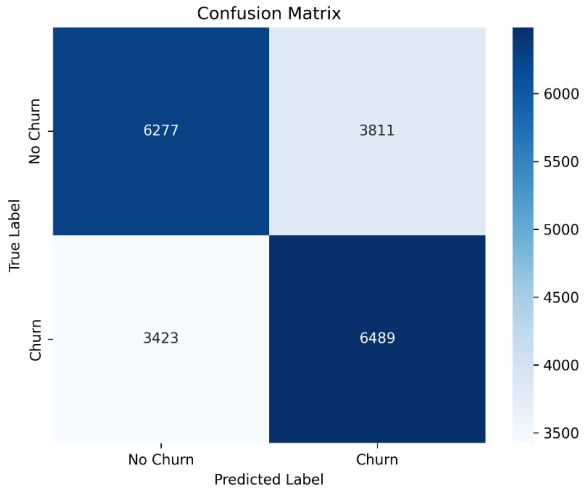

Figure 2: Confusion Matrix for XGBoost model.

Insights into Both evaluation metrics:

The model achieved an AUC-ROC = [0.7] on the holdout test set, confirming the generalization ability established during cross-validation. Specifically, the Recall of [0.65] demonstrates that the model successfully identifies a high proportion of actual customer departures, which is critical for timely intervention.

4.2.7 Model interpretation (SHAP analysis)

The model's complex decision-making process was explained using Shapley Additive exPlanations (SHAP) values. This technique provides local (individual prediction) and global (overall feature importance) explanations for the XGBoost output. It provides:

- **SHAP Summary Plot** → shows how feature values push predictions higher or lower, shown in Figure 3.
- **Feature Importance Ranking (Bar Plot)** → shows which variables influence churn the most, shown in Figure 4.

Top 5 Feature Impact Analysis

The analysis consolidates the impact of the top five features, revealing the primary factors driving and inhibiting customer churn:

1. nu**m_eqpdays (Equipment Age):** The most powerful inhibitor. High values (older equipment) significantly reduce the likelihood of churn, suggesting high customer stability.
2. **num_months (Total Tenure):** High values (long tenure) increase the likelihood of churn, indicating a distinct risk among long-standing accounts not secured by recent contracts.
3. **num_mou_mean (Mean Monthly Usage):** Low usage values increase the probability of churn, serving as a strong indicator of customer disengagement.
4. **num_uniqsusbs (Unique Subscribers in Household):** High values reduce the likelihood of churn, demonstrating that larger, multi-subscriber accounts exhibit greater loyalty and higher switching costs.
5. **num_change_mou (Monthly Usage Change**): Sharp decreases in usage increase the risk, functioning as a critical behavioral signal for imminent churn decisions.

This analysis guides strategy toward proactively rewarding and engaging low-usage or long-tenured customers, while recognizing equipment age and account size as the most reliable factors for stability.

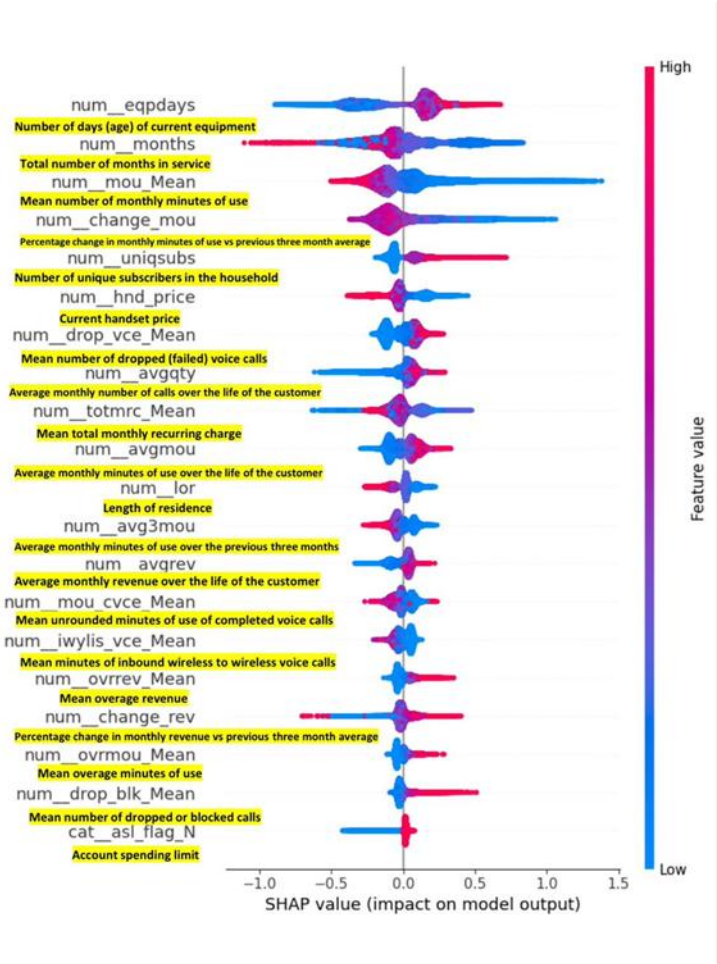

Figure 3: SHAP Summary Plot

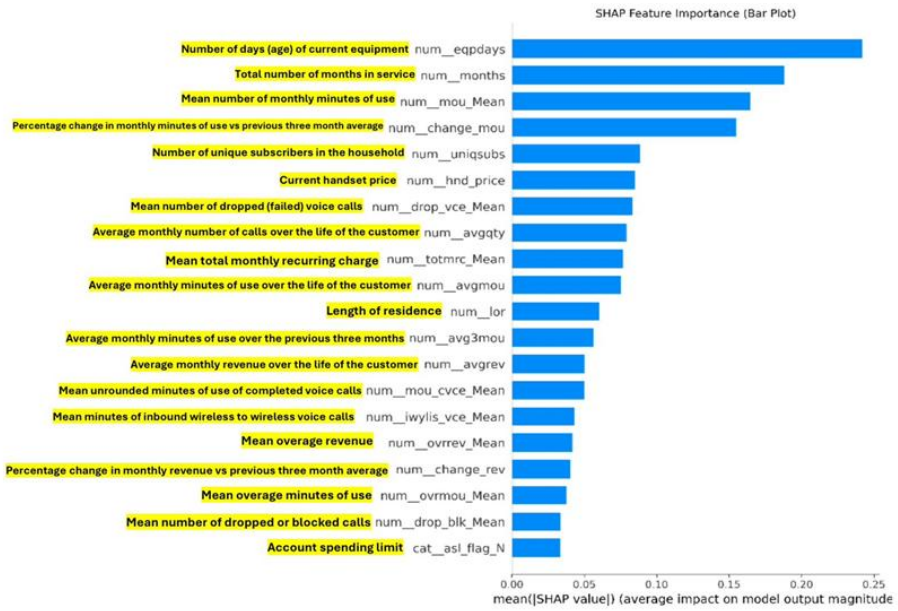

Figure 4: Feature Importance Ranking (Bar Plot)

4.2.8 Retention Strategy

Insights from SHAP analysis provide a clear basis for data-driven, targeted retention strategies. The highest priority should be directed towards the group most at risk, which are customers with low average monthly usage and long service life, as these groups require immediate intervention in the form of personalized re-interaction campaigns or loyalty rewards programs. In contrast, the stability factors indicated by the model, such as older devices and multi-subscriber accounts, can be enhanced through device update offers or family service packages, ensuring that the protective advantage is transformed into a long-term commitment. This focus ensures maximum return on investment (ROI) for customer retention efforts.

4.3 Architecture/Workflow Diagram

To ensure clarity and operational consistency, the predictive system is conceptualized as a modular architecture. This design separates data preparation from the prediction engine, which facilitates seamless integration into existing infrastructure. **Figure 5** illustrates the end-to-end workflow, detailing how raw input data flows through the preprocessing and scoring components to produce actionable outputs.

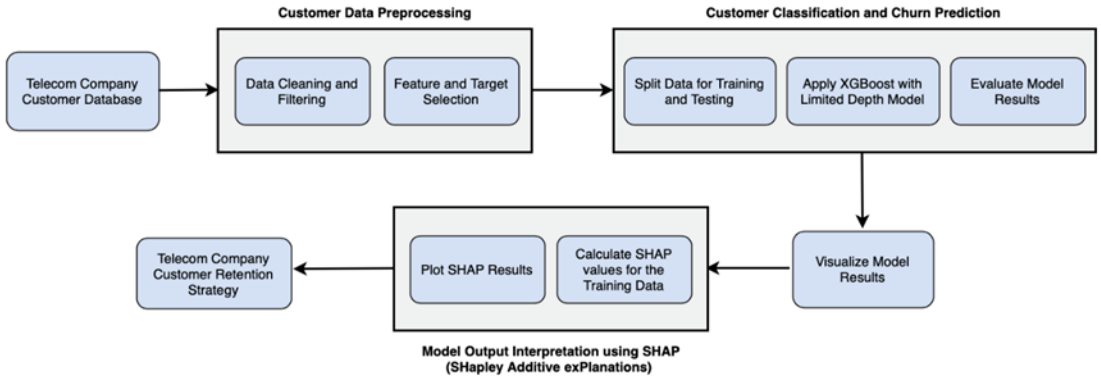

Figure 5 :Architecture/Workflow Diagram

4.4 Integration

The implementation of the predictive module relies on establishing a reliable and consistent production scoring workflow. This process ensures that new customer data is handled identically to the training data before reaching the model for prediction. Table 4 outlines the sequential phase-by-phase flow of data from the raw source to the final actionable trigger within the CRM system.

Phase	Description (What happens?)
Input Data Source	The latest customer data (e.g., call logs, outstanding balances, complaints) is pulled from the Data Warehouse (DW).
Preprocessing & Feature Store	The same ColumnTransformer step and transformations used during the training phase are applied to the new data.
Prediction Endpoint	The processed features are sent to an API Endpoint where the final XGBoost model (***best_xgb_model***) is hosted.
Output Delivery	The model outputs the prediction (e.g., "Churn Probability 0.85") or the binary label ("***Churn***").
Action Trigger	The prediction is sent to the CRM system to be used for an automated action (e.g., placing the customer on the list for preventative campaigns).

Table 4 :Operational Integration Workflow for Real-Time Churn Prediction Scoring

4.5 Expected Benefits

The implementation of the optimized XGBoost model and its surrounding data pipeline is expected to yield several substantial benefits for the telecom operator, focusing on strategic clarity and operational efficiency:

1. ***Systematic Predictive Performance:***

   By utilizing XGBoost tuned via GridSearchCV, the model provides a systematic and reproducible framework for identifying high-risk customers. The achieved AUC-ROC of 0.70 ensures that the predictions offer a reliable probability separation compared to basic baseline methods, enabling effective verification of customers at risk of churn.

2. ***Effective Handling of Class Imbalance:***

   The explicit application of the scale pos weight parameter directly addresses the dataset's inherent imbalance. This is expected to increase the model's Recall (0.65), meaning the model proactively captures a larger proportion of true churn cases, which is critical for successful retention campaigns.

3. ***Enhanced Model Transparency and Trust:***

   Through the rigorous SHAP Analysis, the model transcends the typical opaque "black-box" outputs. It provides quantitative explanations at both global (feature importance) and local (individual customer prediction) levels, building trust with business stakeholders and enabling deeper understanding of the prediction drivers.

4. ***Targeted Business Strategy:***

   The ability to recognize the top drivers of churn (e.g., Num_eqpdays, Num_months) directly translates into actionable business insights. This allows the company to design highly targeted and personalized retention strategies, supporting data-driven decision-making and a more efficient allocation of marketing resources.

5. ***Simplified Operational Integration:***

   The entire modeling process, structured using the Scikit-learn Pipeline and ColumnTransformer framework, facilitates smooth and standardized integration into existing Customer Relationship Management (CRM) systems. This design supports continuous monitoring and real-time prediction deployment without requiring complex system reconstructions.

6. ***Optimized Profitability:***

   The early and reliable detection of churn-prone customers significantly reduces the reliance on costly new customer acquisition efforts. By effectively retaining the existing customer base, the model directly contributes to improved profitability and long-term customer loyalty.

5. CONCLUSION

This project successfully addressed the critical challenge of customer churn prediction within the telecom sector by developing a highly efficient and interpretable predictive module. To achieve this, a hybrid machine learning module was designed using the XGBoost classifier, tuned with GridSearchCV to ensure robust performance, and integrated with SHAP analysis to provide transparent and commercially actionable insights. This methodology successfully identified key churn factors, such as Num_eqpdays and Num_months. The project also efficiently handled the problem of Class Imbalance by applying the scale_pos_weight parameter, a necessary step to boost Recall and mitigate bias.

Despite the success in building an explainable model, the quantitative performance on the test set recorded an AUC-ROC of [0.7] and a Recall of [0.65]. While these results are considered good, they position the model significantly below the advanced accuracy levels (90%-99%) achieved by other ensemble models cited in the literature review. This disparity strongly suggests that the scope or quality of the data used may be the primary constraint preventing the XGBoost model from reaching its full predictive potential, or that the hyperparameter search space was limited. Consequently, future research must focus on seeking larger and more diverse datasets, exploring advanced techniques for handling class imbalance, or adopting alternative models to ensure the churn prediction system achieves the high level of accuracy required to effectively support customer retention decisions.

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